

Muhammad Kamran

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EDUCATION

Bachelor of Science in Computer Science

University of Education (Lahore) **Grade:** 3.24/4.00

SUMMARY

AI/ML Engineer with hands-on experience developing machine learning systems, agentic AI workflows, LLM applications, RAG platforms, computer vision solutions, and production-oriented AI services. Experienced in classical machine learning across regression and classification problems, including data preprocessing, feature engineering, model training, hyperparameter tuning, and performance evaluation.

Strong backend engineering background with Python, FastAPI, REST APIs, PostgreSQL, Docker, and AI service integration. Experienced in designing stateful LangGraph workflows, retrieval systems, structured tool-calling pipelines, human-in-the-loop AI systems, model evaluation frameworks, and scalable backend services for AI-powered applications.

Comfortable working with modern frontend technologies such as Next.js and React when required to deliver complete AI products, while primarily specializing in AI/ML engineering and backend development.

PROFESSIONAL EXPERIENCE

Mid Level AI / ML Engineer – Hubble42 Inc.

Lahore, Punjab, Pakistan · On-site | **Apr 2026 – Present**

- Developed and evaluated **classical machine learning solutions** for regression and classification problems, including preprocessing, feature engineering, model training, hyperparameter tuning, and performance evaluation.
- Designed and evaluated complex Data Science, Software Engineering, and STEM benchmarks for frontier AI models and agentic systems.
- Worked extensively with **agentic AI systems**, analyzing multi-step agent trajectories, tool usage, reasoning failures, and task execution behavior.
- Built end-to-end evaluation datasets, validation pipelines, automated testing workflows, and reproducible task environments for assessing AI model performance.
- Performed detailed quality assurance and failure analysis to distinguish model reasoning errors, infrastructure issues, and task-design problems.
- Used AI development tools including Cursor, Claude Code, Gemini CLI, and GitHub Copilot to accelerate implementation, testing, and validation workflows.

Junior AI / ML Engineer – Hubble42 Inc.

Lahore, Punjab, Pakistan · On-site | **Feb 2026 – Apr 2026**

Junior LLM Engineer – Intelligence Cell, Neodustria

Basel, Switzerland · Remote | **Oct 2024 – Jan 2026**

- Worked on the development and integration of **LLM-based systems** within production workflows, including ontology-grounded AI systems for automotive and construction engineering domains.
- Contributed to scalable AI pipelines and automation systems for enterprise use cases.
- Developed and integrated FastAPI-based services supporting retrieval-augmented generation and LLM inference workflows.
- Assisted in fine-tuning and deployment workflows for **Specialized Engineering Models (SEMs)**.

- Worked with Generative AI technologies including vLLM, LangChain, ontology-based systems, PostgreSQL, GraphDB, and workflow orchestration.
- Assisted with containerized deployments using Docker.
- Contributed to 3D auto-labeling pipelines and dataset preparation workflows for AI model training using PointNet++ and mesh-processing techniques.
- Collaborated on AI research, experimentation, and system optimization tasks.

RESEARCH EXPERIENCES

AI/ML Intern – National Center of Artificial Intelligence (NCAI)

University of Engineering & Technology (UET), Lahore | **May 2024 – Sep 2024**

- Contributed to a research-based traffic congestion detection system under senior guidance.
- Implemented machine learning and deep learning models across multiple AI projects.
- Worked on data annotation, preprocessing, and dataset preparation for model training.
- Supported computer vision workflows, including preparing and feeding annotated data into AI models.
- Assisted with model training, experimentation, and evaluation to improve project outcomes.

PROJECTS

ScopeAI – Agentic Client Discovery & Project Planning Platform

LangGraph | **Python** | **FastAPI** | **Supabase** | **PostgreSQL** | **Next.js** | **React** | **TypeScript**

- Built an **agentic AI platform** that transforms unstructured client briefs into structured requirements, clarification questions, solution architectures, risk analyses, and implementation roadmaps.
- Designed a **stateful LangGraph workflow** using specialized processing nodes, conditional routing, structured outputs, controlled tool execution, checkpointing, and human-in-the-loop interactions.
- Developed FastAPI-based backend services for AI workflow execution, project context management, LLM integration, and communication between the agent layer and application data services.
- Implemented controlled tool interfaces for retrieving and updating project information while separating probabilistic LLM reasoning from deterministic validation and database operations.
- Integrated Supabase/PostgreSQL for authentication, project persistence, workflow data, file storage, and Row Level Security.
- Added workflow observability for tracking agent execution stages, tool calls, interruptions, failures, and generated artifacts.
- Built a Next.js/React interface to expose the AI workflow through project dashboards, clarification forms, requirements review, risk analysis, and roadmap views.

Ontology-Grounded RAG Platform

Python | **FastAPI** | **LangChain** | **PostgreSQL** | **Docker**

- Developed **retrieval-augmented generation workflows** for engineering knowledge systems.
- Built FastAPI services for document ingestion, semantic retrieval, structured knowledge access, and LLM inference.
- Integrated vector retrieval with structured and ontology-driven knowledge sources.
- Containerized services using Docker and participated in deployment, testing, and performance validation.

Context-Aware Chatbot for Freedomz Storage

Generative AI | **LangChain** | **Groq API** | **FastAPI** | **React** | **Hugging Face** | **ChromaDB** | **WordPress**

- Designed and developed a full-stack **RAG chatbot** for a real-world storage business.
- Built document retrieval using Hugging Face embeddings, ChromaDB, chunking strategies, and semantic similarity search.

- Developed the FastAPI backend and React frontend and integrated the application with an existing WordPress website.
- Implemented prompt and retrieval controls to keep responses grounded in business-specific information.

FitAI – AI-Powered Fitness Companion

Computer Vision | OpenCV | MediaPipe | FastAPI | LangChain | Hugging Face

- Developed a **computer vision system** for real-time exercise posture and movement analysis.
- Extracted body keypoints and joint-angle features to identify incorrect exercise patterns.
- Integrated AI-generated workout guidance and conversational assistance.
- Built backend services for exercise analysis, recommendations, and AI-powered user feedback.

AI Automation Workflow System

n8n | LLM APIs | Workflow Automation | API Integration

- Designed AI-powered automation workflows connecting LLM APIs with external services.
- Built multi-step pipelines for data processing, content generation, transformation, and structured output delivery.
- Automated repetitive operational workflows through event-based orchestration and API integrations.

Bone Fracture Detection

YOLO | Computer Vision | OpenCV | Deep Learning

- Developed an object-detection model for identifying bone fractures in X-ray images.
- Prepared and annotated medical-imaging datasets and performed image preprocessing.
- Trained and evaluated YOLO-based detection models for healthcare-related computer-vision tasks.

Web Scraping & ETL Pipelines

Python | Pandas | SQL | PostgreSQL

- Developed web-scraping pipelines for extracting structured data from 100+ sources.
- Designed ETL workflows for cleaning, transforming, validating, and storing datasets.
- Prepared structured datasets for analytics and downstream machine-learning applications.

SOFTWARE AND PROGRAMMING SKILLS

- **Programming:** Python, SQL, TypeScript, JavaScript
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, PyTorch, XGBoost, Regression, Classification, Feature Engineering, Model Training, Hyperparameter Tuning, Model Evaluation
- **Generative AI & Agentic Systems:** LangGraph, LangChain, RAG, LLM Applications, Tool Calling, Stateful Agents, Human-in-the-Loop AI, Structured Outputs, Prompt Engineering, Hugging Face, vLLM, LLM Evaluation
- **Backend Engineering:** FastAPI, REST APIs, API Design, Microservices, Authentication, API Integration, Backend Services
- **Databases & Retrieval:** PostgreSQL, Supabase, SQLite, ChromaDB, GraphDB, Vector Search
- **Computer Vision:** OpenCV, YOLO, MediaPipe, Pose Estimation, PointNet++
- **Data Engineering:** Pandas, NumPy, ETL Pipelines, Data Preprocessing, Web Scraping
- **DevOps & AI Infrastructure:** Docker, Git, GitHub, Vercel, Hugging Face Spaces, Prefect, Workflow Automation
- **Frontend:** Can work using AI coding assistants
- **AI Development Tools:** Cursor, Claude Code, GitHub Copilot, Gemini CLI

CERTIFICATIONS

- Microsoft Certified: Azure AI Fundamentals
- NAVTTC Certified Artificial Intelligence Course
- Machine Learning – Andrew Ng (Coursera)
- Python for Everybody (Coursera)